Quantum Computing, Grover's Algorithm, and Introduction to PQC

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Quantum Computer

Quantum Computer

A computer that uses principles of quantum mechanics for computation.

Misconception

Faster in specific problems. Not faster in general problems.

Warning

This is a very brief description of quantum computing.

Quantum Mechanics

State

A state $|\phi\rangle$ of quantum system is an element of complex inner product space $\mathcal{H}$. Assume that $\| |\phi\rangle \| = 1$.

Schrödinger Equation (Time-Dependent)

$$i\hbar \frac{\mathrm{d}}{\mathrm{d}t} |\Psi\rangle = \hat{H} |\Psi\rangle$$

Schrödinger Equation (Time-Independent)

$$H |\psi\rangle = E |\psi\rangle$$

where E is energy (constant).

State Vector

Qubit

Qubit is a quantum bit where it is a superposition of two states.

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

where $|\alpha|^2 + |\beta|^2 = 1$, $\alpha, \beta \in \mathbb{C}$

Born Interpretation

Given state vector $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, at observation:

$$P(|0\rangle) = |\underbrace{\langle 0|\psi\rangle}_{\text{Extract 0-coeff}}|^2 = |\alpha|^2$$

$$P(|1\rangle) = |\underbrace{\langle 1|\psi\rangle}_{\text{Extract 1-coeff}}|^2 = |\beta|^2$$

Observation

Once observed, state vector of qubit "collapses" such that the probability distribution is 1 for the observed state, and 0 for others.

```
class Qubit {
    public:
    Qubit(const ComplexFloat alpha) : alpha(alpha) {}

    State observe() {
        float p[2]; //probability
        p[0] = abs(alpha) * abs(alpha);
        p[1] = 1 - p[0];

        State ret = chooseByProb(p);

        // Collapse (Ignore phase)
        alpha = (ret == 0) ? 1.0 : 0.0;
        return ret;
    }
    private:
    ComplexFloat alpha;
};

```

Example

Suppose a qubit has the following state vector:

$$|\psi\rangle = \frac{1}{2} i\,|0\rangle - \frac{\sqrt{3}}{2}\,|1\rangle$$

Suppose at "observation", the qubit held the state "1". Immediately afterwards,

$$|\psi\rangle = e^{i\theta}\,|1\rangle$$

where θ is a random phase.

Multi-qubit System

2-qubit Computer

Suppose a computer has two qubits, then one could write:

$$|\psi\rangle = \alpha\,|00\rangle + \beta\,|01\rangle + \gamma\,|10\rangle + \delta\,|11\rangle = \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}$$

where $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$.

Remark

An n qubit-system can represent 2^n states.

Quantum Gate

Classical Gates

In classical computers, we have:

- not
- or
- and
- xor

and these are combined to create a computer.

Quantum Gates

In quantum computers, any unitary operator/matrix H acting on state vector $|\psi\rangle$ is a quantum gate, that is,

$$H^* H = \left(\bar{H}\right)^T H = I$$

Example: Hadamard Gate

For 1-qubit state $|\psi\rangle$,

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

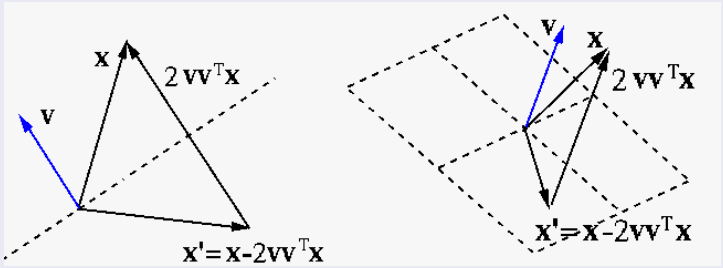
So that,

$$H|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$
$$H|1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

Example: "Householder"

Suppose $|\phi\rangle$ is a separate state vector (note that it is unit vector). For n-qubit (2^n-state system), define H as

$$H = I - 2\,|\phi\rangle\,\langle\phi|$$



Quantum Oracle

Classical Oracle

Function f such that

$$f(x = password) = 1$$
$$f(x \neq password) = 0$$

Quantum Oracle

Quantum operator such that, for specific "secret state" w,

$$H|w\rangle = -|w\rangle$$
$$H|\psi\rangle = |\psi\rangle$$

Example: 2-qubit Password

The password is $|01\rangle$. The quantum oracle for this is:

$$H = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

So that,

$$\begin{aligned} H|00\rangle &= |00\rangle \\ H|01\rangle &= -|01\rangle \\ H|10\rangle &= |10\rangle \\ H|11\rangle &= |11\rangle \end{aligned}$$

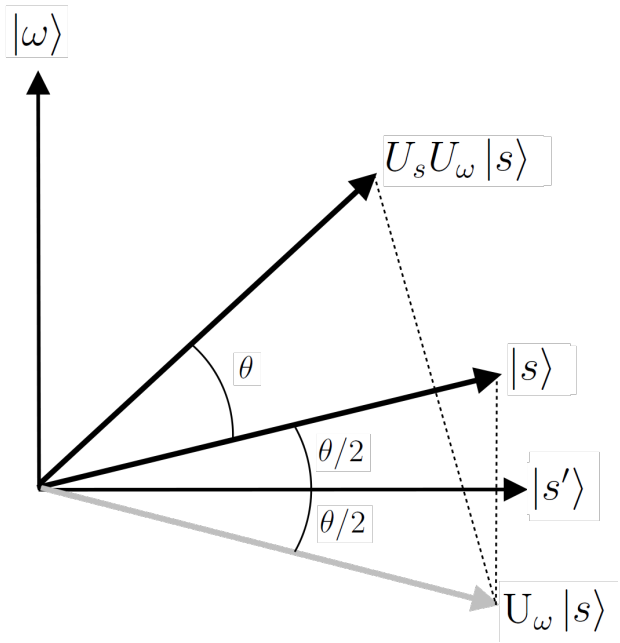
Grover's Algorithm

Grover's Algorithm

Quantum algorithm for breaking an oracle.

Suppose F is a quantum oracle.

1. Apply Hadamard gate to each qubit of the quantum system for initialization.
   - $|s\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$
2. Repeat the following two steps until satisfied.
   1. Apply F to the system.
   2. Apply $2|s\rangle\langle s| - I$ to the system (Augmented Householder)
3. Observe the qubits. Note that this observation "resets" the system.



where

- U_ω is the quantum gate for the oracle with password ω.
- $U_s = 2|s\rangle\langle s| - I$ where s is initialized state.
- At iteration i, the state vector is $(U_s U_\omega)^i |s\rangle$.
- After long iteration, state vector is almost parallel to $|\omega\rangle$.
- $\theta = 2 \arcsin \frac{1}{\sqrt{N}}$, so to travel about $\frac{\pi}{2}$, it takes about $\frac{\pi}{4 \arcsin 1/\sqrt{N}} = O\left(\sqrt{N}\right)$.

Optimal Running Time

For Grover's algorithm, for near-optimal measurement, use $\frac{\pi}{4}\sqrt{2^n}$ iterations for n-qubit system over 2^n potential solutions.

Warning

Grover's algorithm **cannot** be resumed after inspection; you have to start over.

Demonstration

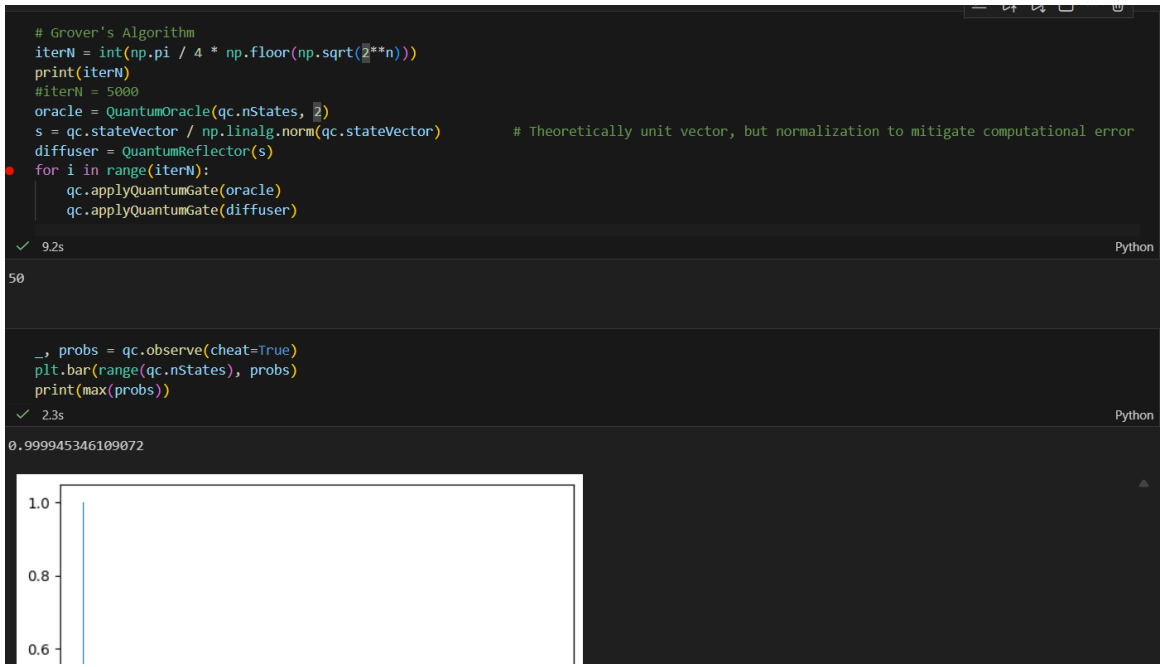


Figure: Grover's Algorithm on 12-qubit system for solving a random oracle over $2^{12} = 4096$ inputs. 50 iterations needed.

Caveats

Big Issues

- Qubit Error
  - Due to environmental noise, etc.
  - Robust qubit "logical qubit" conceptualized with Quantum Error Correction (QEC).
- Quantum Circuit for Oracle Needed.

Post-Quantum Cryptography

Classical Asymmetric Algorithms

- RSA
- ECC

These rely on difficulty of factorization and discrete logarithm problems. (Potentially can be defeated by Shor's algorithm.)

PQC

- Lattice-based Cryptography
- Multivariate Cryptography
- etc.

ML-KEM

ML-KEM

Module-**L**attice-Based **K**ey-**E**ncapsulation **M**echanism

- Shared Secret Generation (but not Diffie-Hellman)
- Lattice-based Cryptography

NIST Standards

- FIPS 203
- Originally CRYSTALS-Kyber (Star Wars)

ML-DSA

ML-DSA

Module-**L**attice-Based **D**igital-**S**ignature **A**lgorithm

- Digital signature
- Lattice-based Cryptography

NIST Standards

- FIPS 204
- Originally CRYSTALS-Dilithium (Star Trek)

SW > SW

아우토크립트, 양자 암호기술 대응 차세대 솔루션 'AutoCrypt PKI-Vehicles' 출시

발행일 : 2025-12-08 08:53 지면 : 2025-12-09 22면



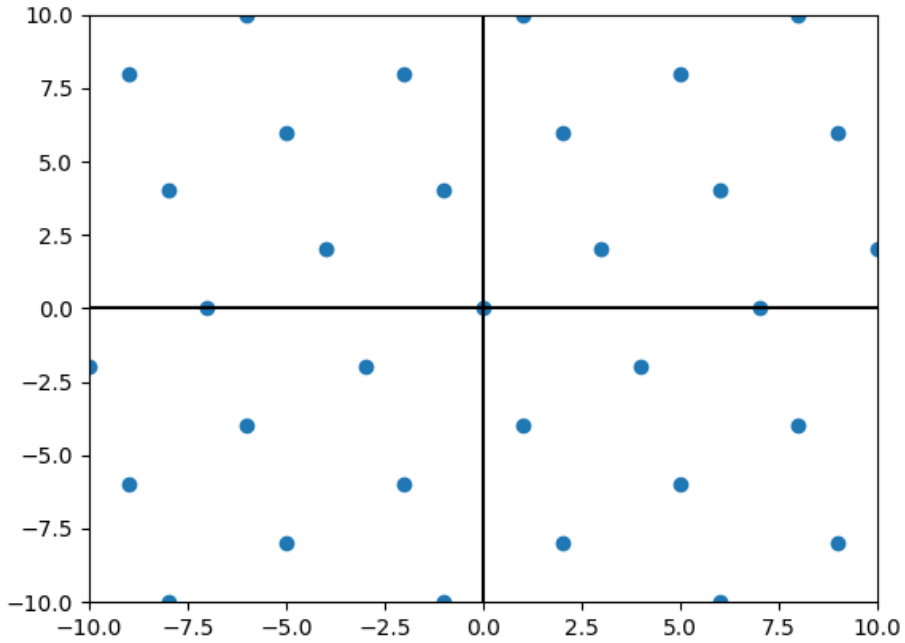
AI 모빌리티 보안기업 아우토크립트(공동대표 이석우·김덕수)는 양자내성암호(PQC) 알고리즘 ML-DSA를 지원하는 국내 최초 상용 인증·인프라 솔루션인 'AutoCrypt PKI-Vehicles'를 출시했다고 8일 밝혔다.

아우토크립트는 차세대 암호 기술이 적용된 솔루션을 국내 최초로 선보임으로써 급성장하는 AI 모빌리티 보안 시장에서 기술 표준을 선도하는 입지를 확고히 하게 됐다.

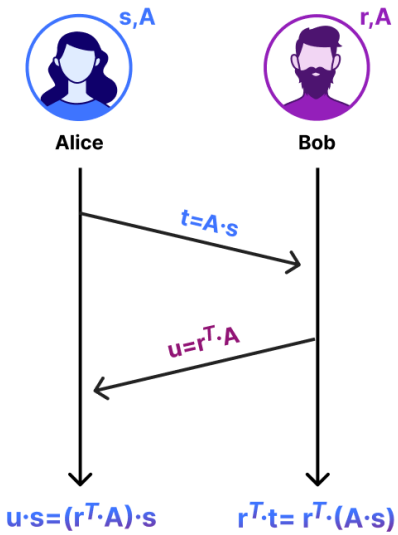
Lattice-based Cryptography

Lattice

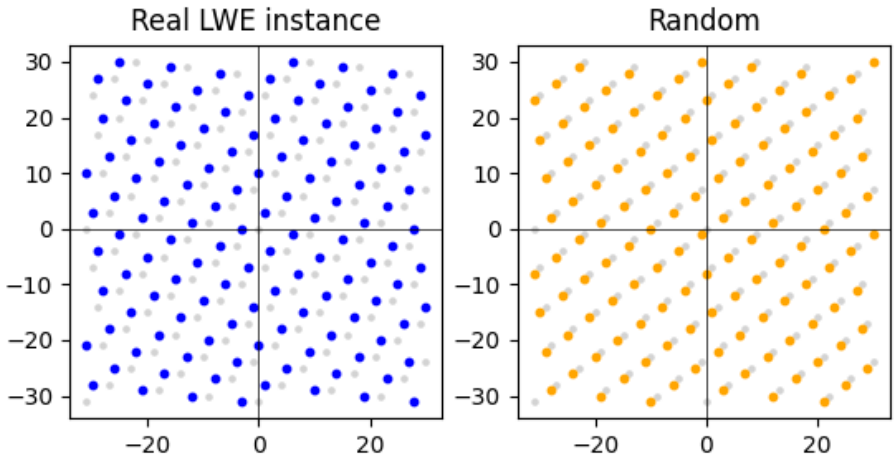
Given basis $\{b_1, \cdots, b_n\}$, lattice $L = \{\sum a_i b_i | a_i \in \mathbb{Z}\}$



ML-KEM Motivation



By knowing A and t, one could figure out s.
Instead, use $t = As + \underbrace{e}_{\text{Error term}}$



Distinguishing random lattice from lattice with "engineered error" can be difficult to distinguish. Solving this requires solving "shortest vector problem", an NP problem.